Optimization in Octave/Matlab

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Optimization intro

For a given $f\colon [a,b] \to \mathbb{R}$ we search for a point $\xi \in (a,b)$ for which

$$f(\xi) \leq f(x)$$

if $|x-\xi|$ is small enough. We say that $f(\xi)$ is a local minimum and ξ is a local minimizer. We know from calculus that

$$f'(\xi) = 0$$

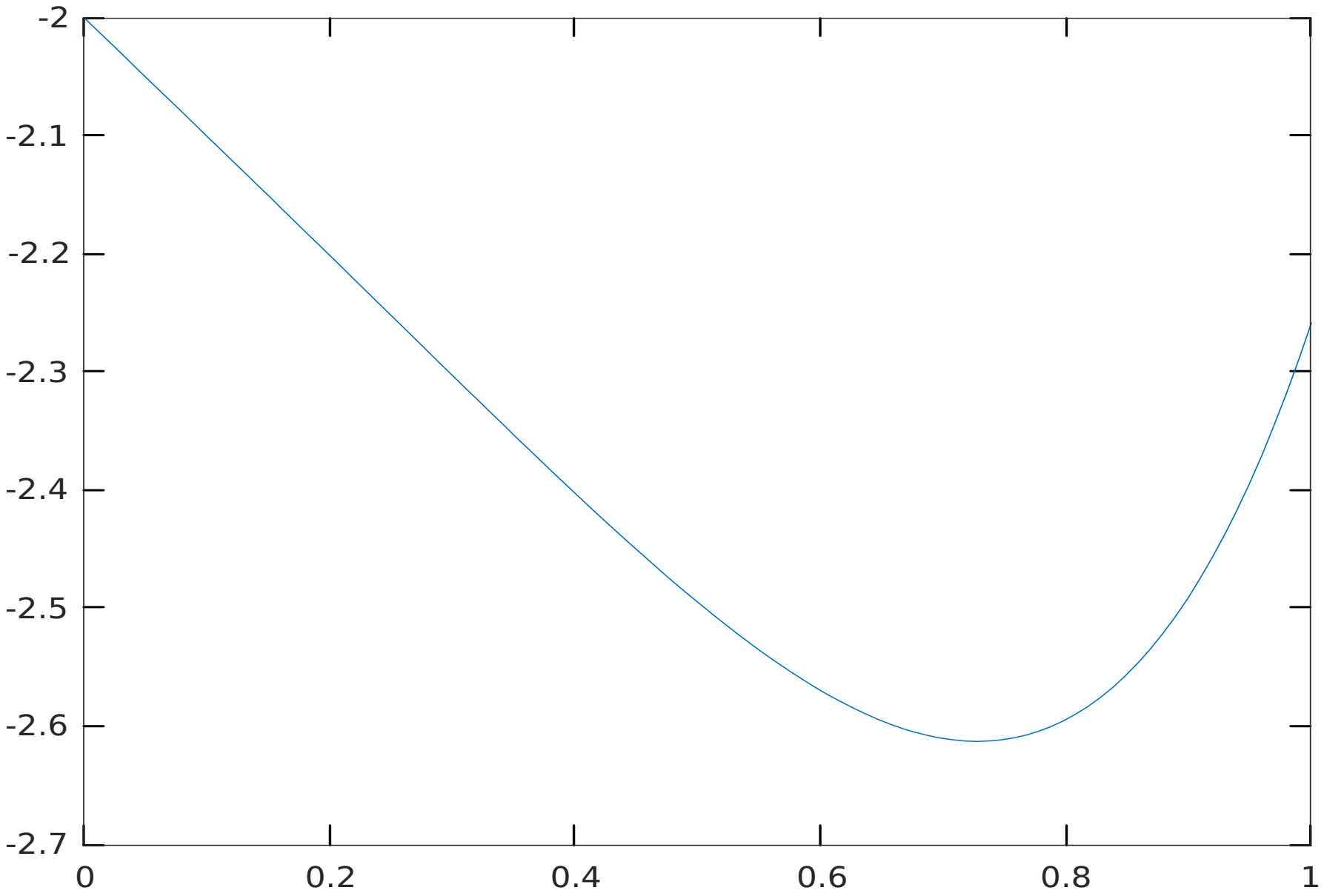
at a local minimizer.

Example 1

Find a local minimizer for

$$\boxed{f(x) = x^5 - cos(x) - e^x}$$

on $[0,1]$ with the Newton method. First, we examine the plot of it:


-2
-2.1
-2.2
-2.3
-2.4
-2.5
-2.6
-2.7
0
0.2
0.4
0.6
0.8
1

Now it is clear that our function has a local minimizer on $[0, 1]$. After computing the derivatives:

$$\boxed{f'(x) = 5x^4 + sin(x) - e^x \text{ and } f''(x) = 20x^3 + cos(x) - e^x}$$

we can easily approximate the root of f', which is a minimizer of f with Octave/Matlab:

```
>> f=@(x) x.^5-cos(x)-exp(x);
>> df=@(x) 5*x.^4+sin(x)-exp(x);
>> ddf=@(x) 20*x.^3+cos(x)-exp(x);
>> x=0.7;
>> x=x-df(x)/ddf(x)
x =  0.730125169245811
>> x=x-df(x)/ddf(x)
x =  0.728187065551714
>> x=x-df(x)/ddf(x)
x =  0.728178496917075
>> x=x-df(x)/ddf(x)
x =  0.728178496750211
```

In Octave/Matlab, we have two possibilities for computing the zeros:

The function `fzero` can be used for univariate functions. By default, it needs an initial bracketing (an interval that consists the zero), but can be used also without specifying it:

```
>> fzero(df, [0, 1])
ans =  0.728178496750212
>> fzero(df, 0.6)
ans =  0.728178496750211
```

The function `fsolve` can be applied for multivariate functions. Here we only have to specify an initial guess of the zero:

```
>> fsolve(df, 0.6)
ans =  0.728178501057600
```

As we see, the results can be differ slightly. Their behaviour can be fine tuned, through setting appropriate parameters, for details see the help.

Example 2

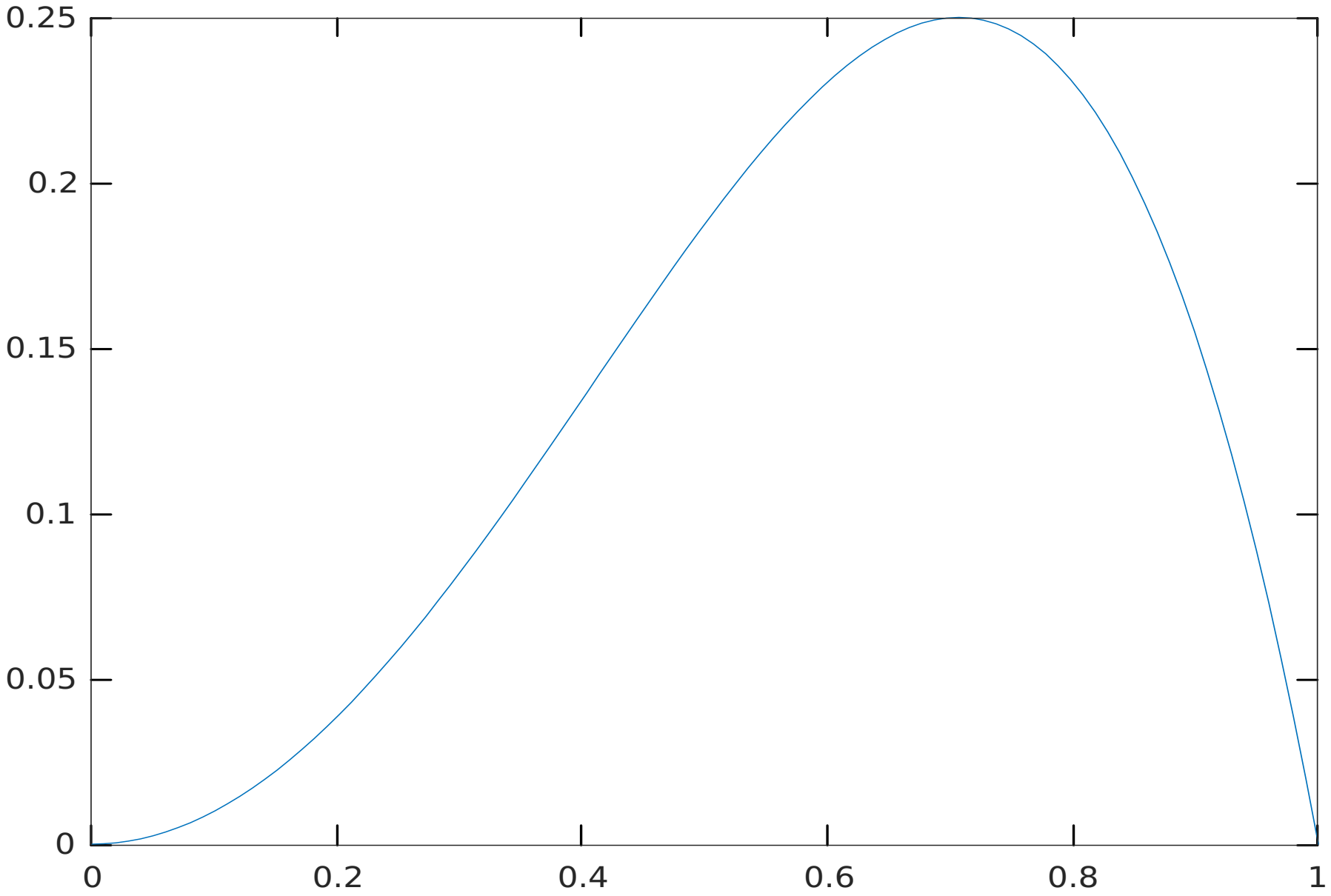
Of all right angled triangle with a given hypotenuse,find that whose area is the greatest. Denote the length of the hypotenuse by c and by x one of the other sides. Applying the Pythagorean theorem we get $\sqrt{c^2 - x^2}$ for the third side, therefore the area of our triangle is

$$A(x) = 0.5x\sqrt{c^2 - x^2}$$

where $0 < x < c$. Now, in order to avoid the difficulties of computing the derivatives, we omit the factor 0.5 and take square of the remaining expression:

$$(2 * A(x))^2 = B(x) = x^2(c^2 - x^2) = c^2x^2 - x^4$$

Let us examine the plot of B with $c = 1$:


0.25
0.2
0.15
0.1
0.05
0
0
0.2
0.4
0.6
0.8
1

Apply again the Newton method and compare the result with the output of `fzero`:

```
>> B=@(x,c) c^2*x.^2-x.^4 ;
>> dB=@(x,c) 2*c^2*x-4*x.^3 ;
>> ddB=@(x,c) 2*c^2-12*x.^2 ;
>> c=1; x=0.7;
>> x=x-dB(x,1)/ddB(x,1)
x =  0.707216494845361
>> x=x-dB(x,1)/ddB(x,1)
x =  0.707106806711824
>> x=x-dB(x,1)/ddB(x,1)
x =  0.707106781186549
>> fzero(@(x)dB(x,1),0.7)
ans =  0.707106781186548
```

Optimization with built-in functions

There is a simpler way to find a minimizer for a univariate function, we can apply the function `fminbnd`. We pass a function and an interval, and `fminbnd` tries to find the minimal value on the given interval:

```
>> B=@(x) -x.^2 + x.^4 ;
>> [xopt, fopt]=fminbnd(B,0.1,1)
xopt =  0.707106781164670
fopt = -0.250000000000000
```

Note that we changed the sign of the function B in order to get maximization problem.

Exercise 1

Find a local minimizer for

$$f(x) = x^5 - cos(x) - e^x$$

using `fminbnd`.

Exercise 2

Find a local minimizer for

$$f(x) = x^x - cos(x) - 0.1 \cdot e^x$$

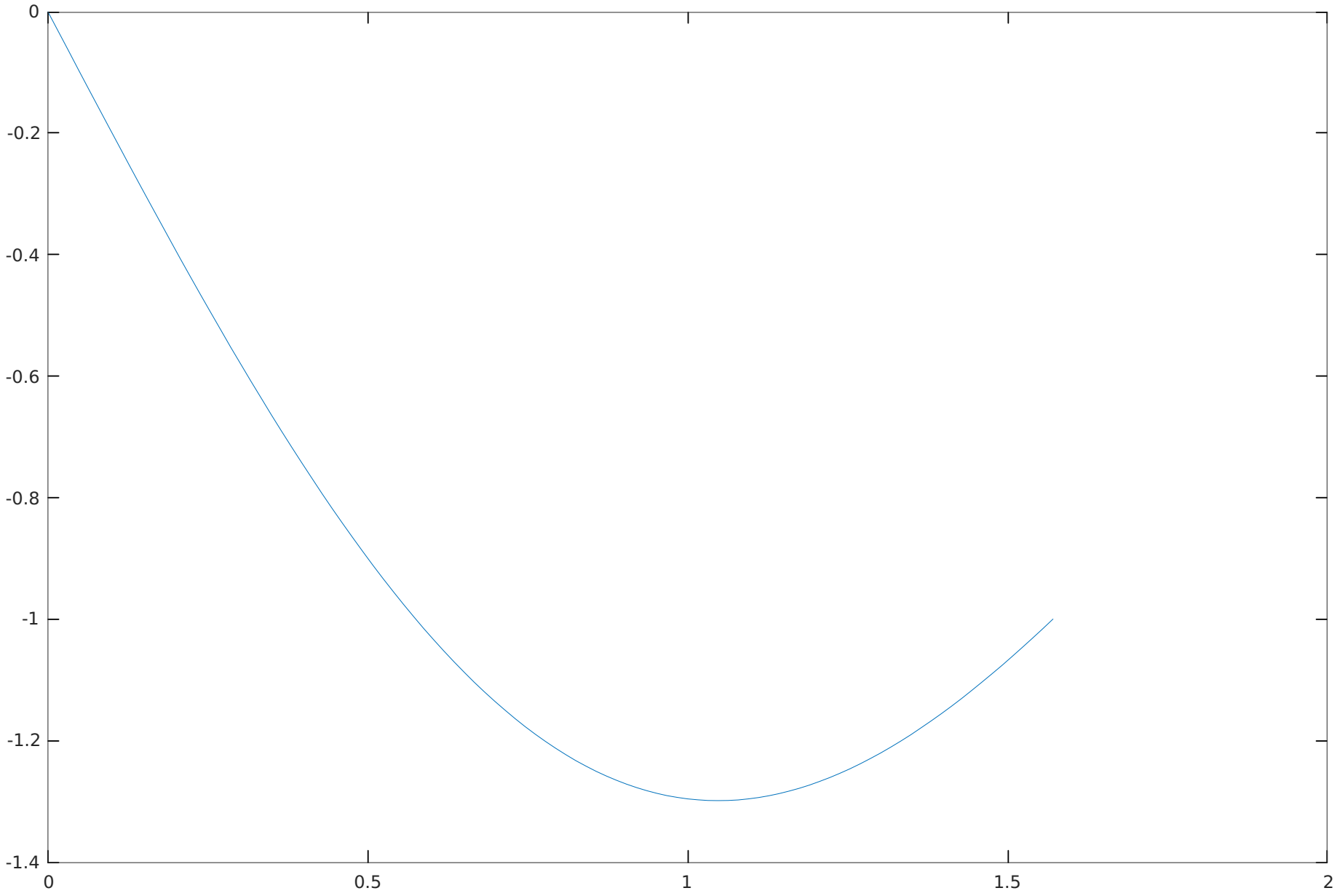
using `fminbnd`.

Example 3

Find the maximizer of $\sin(\alpha)(1+\cos(\alpha))$ on $[0, \frac{\pi}{2}]$!

Solution:

```
f= @(x) -sin(x).*(1+cos(x)); # maximizing -> negative sign
xx=linspace(0,pi/2);
clf;
plot(xx,f(xx));
m2=fminbnd(f,0,pi/2);
m3=fminsearch(f,pi/4);
m,m2,m3
```


0
-0.2
-0.4
-0.6
-0.8
-1
-1.2
-1.4
0
0.5
1
1.5
2

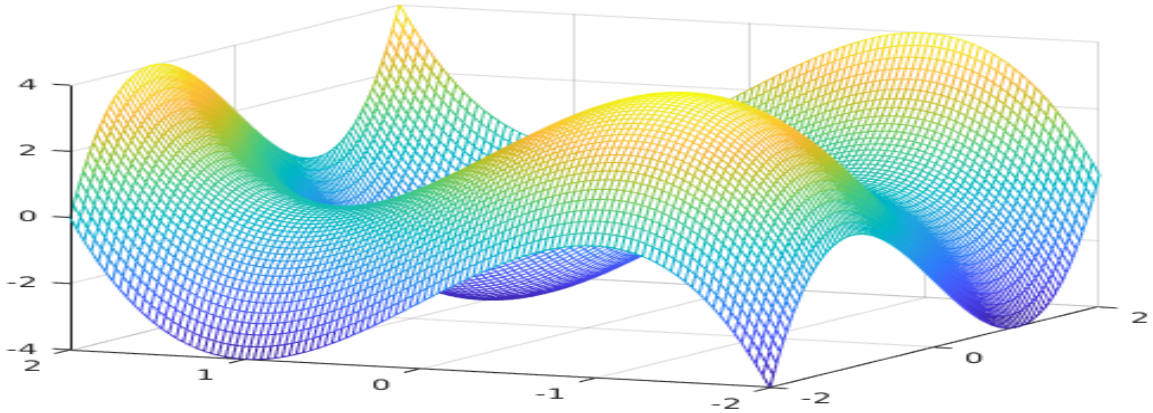
Optimization of multivariate functions

We look for the local minimizers of the function:

$$f(x_1, x_2) = x_1^3 + x_2^3 - 3x_1 - 3x_2$$

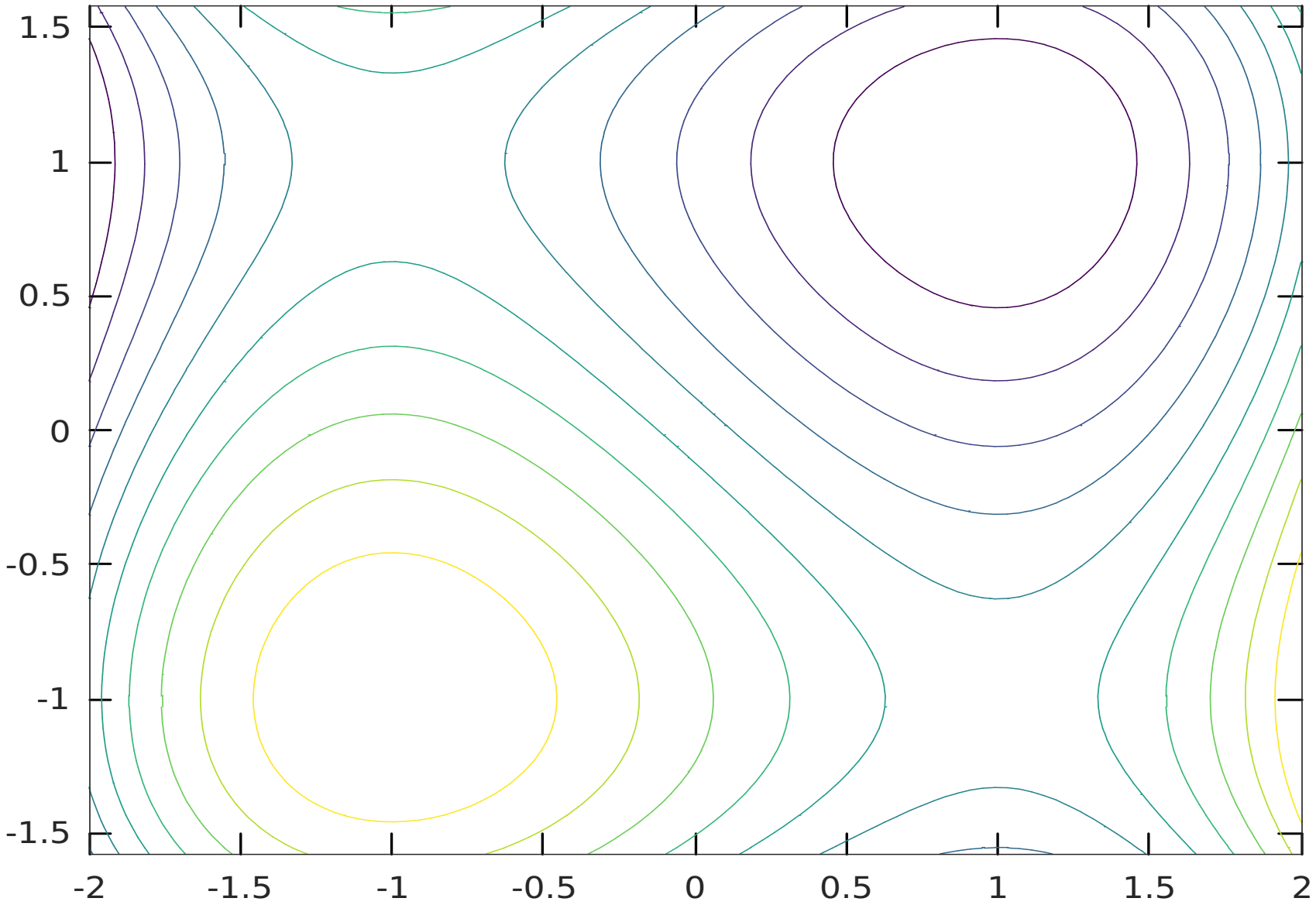
on $[-2, 2] \times [-2, 2]$
To get an overview of a function, we create its plot:

```
>> xx=linspace(-2,2);
>> [X,Y]=meshgrid(xx,xx);
>> Z=X.^3+Y.^3-3*X-3*Y;
>> mesh(X,Y,Z)
```


4
2
0
-2
-4
2
1
0
-1
-2
-2
0
2

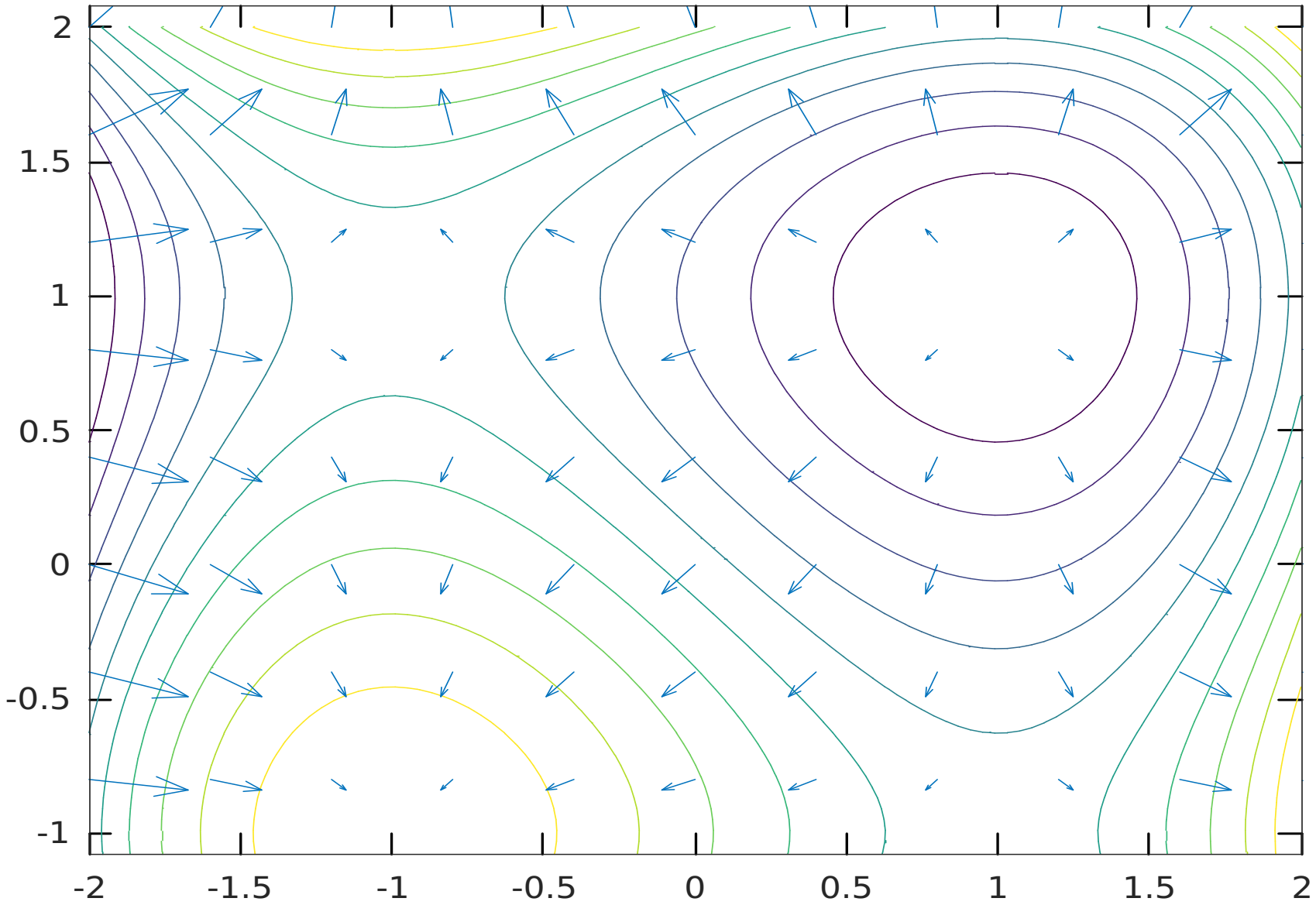
The optimization methods often need an initial guess to start the process. You can guess good starting points, vectors by examining the contour plot. https://en.wikipedia.org/wiki/Contour_line

```
xx=linspace(-2,2);
[X,Y]=meshgrid(xx,xx);
Z=X.^3+Y.^3-3*X-3*Y;
contour(X,Y,Z)
axis equal
```


1.5
1
0.5
0
-0.5
-1
-1.5
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2

Another useful plot for $\mathbb{R}^2 \to \mathbb{R}$ functions is a gradient plot.
https://en.wikipedia.org/wiki/Gradient
Its most important property: the gradient at (x_1, x_2) has the direction of greatest increase of the function at (x_1, x_2).
The corresponding Octave/Matlab code:

```
xx=linspace(-2,2,11);% less points for the gradients
[X,Y]=meshgrid(xx,yy);
dX=3*X.^2-3;
dY=3*Y.^2-3;
hold on; quiver(X,Y,dX,dY)
```


2
1.5
1
0.5
0
-0.5
-1
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2

If the gradient is not available, or hard to compute, one can use the builtin `gradient` function:

```
xx=linspace(-2,2,11);
[X,Y]=meshgrid(xx,yy);
Z=X.^3+Y.^3-3*X-3*Y;
[dX,dY]=gradient(Z);
hold on; quiver(X,Y,dX,dY)
```

optimization with built-in functions

The function `fminsearch` uses derivative-free method to approximate the local minimizer. The parameters: the function handle and the initial guess:

```
>> f=@(x) x(1)^3+x(2)^3-3*x(1)-3*x(2);
>> [xo,fxo]=fminsearch(f,[0.5,0.5])
xo =

   0.999961405649954   1.000002133792563

fxo = -3.99999999551783
```

optimization with built-in functions

The function `fminunc` uses derivative-based method to approximate the local minimizer. The parameters: the function handle and the initial guess:

```
>> f=@(x) x(1)^3+x(2)^3-3*x(1)-3*x(2);
>> [xo,fxo]=fminunc(f,[0.5,0.5])
xo =

   1.00000000468604   1.00000000468604

fxo = -4.00000000000000
```

Meshgrid,contour,mesh,gradient

See the additional slides for optimization on the e-learning site!

Exercise 3

Plot the function, its contour-lines and the gradient field on the given domain. Approximate a local minimizer-maximizer!

$$f(x_1, x_2) = \frac{x_1^3}{6} - x_1 + \frac{x_1 x_2^2}{4} \quad \text{if } (x_1, x_2) \in [-2.6] \times [-2.6]$$

Exercise 4

Plot the function, its contour-lines and the gradient field on the given domain. Approximate a local minimizer-maximizer!

$$f(x_1, x_2) = \sin(x_1)\cos(x_2) \quad \text{if } (x_1, x_2) \in [0, \pi] \times [0, \pi]$$

Exercise 5

Plot the Rosenbrock-function, its contour-lines and the gradient field. Approximate a local minimizer-maximizer!
`https://en.wikipedia.org/wiki/Rosenbrock_function`